Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/08/2020
PI Name:	Hirsa, Amir Ph.D.		
Project Title:	Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hirsaa@rpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	518-269-2817
Organization Name:	Rensselaer Polytechnic Institute		
PI Address 1:	Mechanical, Aerospace & Nuclear Engineering		
PI Address 2:	110 8th St, JEC-2049		
PI Web Page:			
City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	09/30/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:	8	**No. of Master' Degrees:**	4
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	20	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
Contact Email:	sridhar.gorti@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2020 per PI and NSSC information (Ed., 6/4/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lopez, Juan Ph.D. (Arizona State University)		
Grant/Contract No.:	NNX13AQ22G		
Performance Goal No.:			
Performance Goal Text:			

Formation of amyloid fibrils is widely studied because of their role in neurodegenerative diseases, including Alzheimer's. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection due to gravity on the fibrillization process. We know that fluid interfaces (e.g., air-water interface) induce amyloid fibrils when they are sheared, but the influence of

Task Description:	container walls is not fully understood. Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation by removing other effects, including solid walls and buoyancy effects. Formation of protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container, since surface tension can provide liquid containment. Thus, the free surface can be readily observed (e.g., with high magnification objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a monomolecular oil film). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops in an experiment referred to as the Ring-Sheared Drop (RSD).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they lose their native confirmation (folding) and then can bond with other proteins, forming an aggregate. This is the reason why the transparent fluid in an egg becomes a white solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very intriguing because they are highly ordered and self-assemble. When heated, insulin molecules, for example, denature and bond with other denatured molecules (or assists in denaturing the other molecules). These form a nucleation site, and from there the fibril elongates as more molecule bond into what is called a "beta sheet" structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow friction, aka "shear stress," and interfacial effects that native proteins are exposed to. Aside from their central role in neurodegenerative diseases such as Parkinson's and Alzheimer's, formation of amyloid structures in proteins plays a role in other diseases including diabetes.
Task Progress:	Significant progress was made this year both toward basic science as well engineering in support of the flight hardware and preparations for the experiment aboard the ISS. Contributions to basic science include experimental and computational models for flow in the Ring-Sheared Drop (RSD). Also, new experiments were conducted that showed a large degree of protein amyloid fibril interaction with solid surfaces, providing additional incentive for the containerless mixing made possible by the RSD system. Contributions to engineering effort in support of Teledyne-Brown Engr. ("TBE," a NASA contractor) in finalizing the space hardware include material testing to determine contact angle of knife edges as well as protein fibrillization studies in an Earth-based analog of the RSD (known as the knife-edge surface viscometer, or KEV). We were able to eliminate some candidate materials, including stainless steel as ring material due to deterioration. Contributions toward flight preparations include significant modifications to protein sample degassing and freezing protocol proposed by TBE to minimize deleterious effects of freezing on protein sample activity upon thawing.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Abstracts for Journals and Proceedings	Middlestead H, Debono N, Raghunandan A, Hirsa A. "Interfacial-shear induces protein amyloid fibrils." Presented at the 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Seattle, WA, November 23-26, 2019. Bulletin of the American Physical Society. 2019 Nov;64(13):M03.00036. http://meetings.aps.org/Meeting/DFD19/Session/M03.36 , Nov-2019
Abstracts for Journals and Proceedings	Griffin S, Riley F, Gulati S, Hirsa A, Lopez J. "Experiments in oil-water ring-sheared drop." Presented at the 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Seattle, WA, November 23-26, 2019. Bulletin of the American Physical Society. 2019 Nov;64(13):Q25.00001. https://meetings.aps.org/Meeting/DFD19/Session/Q25.1 , Nov-2019
Abstracts for Journals and Proceedings	Adam JA, Gulati S, Bonocora RP, Hirsa AH. "Growth of microorganisms in a space-bioreactor analog." Presented at the 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Seattle, WA, November 23-26, 2019. 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Seattle, WA, November 23-26, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Raghunandan A, Hirsa AH, Underhill PT, Lopez JM. "Predicting steady shear rheology of condensed-phase monomolecular films at the air-water interface." Phys Rev Lett. 2018 Oct 19;121(16):164502. https://doi.org/10.1103/PhysRevLett.121.164502 ; PubMed PMID: 30387637 , Oct-2018
Articles in Peer-reviewed Journals	Gulati S, Riley FP, Hirsa AH, Lopez JM. "Flow in a containerless liquid system: Ring-sheared drop with finite surface shear viscosity." Physical Review Fluids. 2019 Apr;4(4):044006. https://doi.org/10.1103/PhysRevFluids.4.044006 , Apr-2019